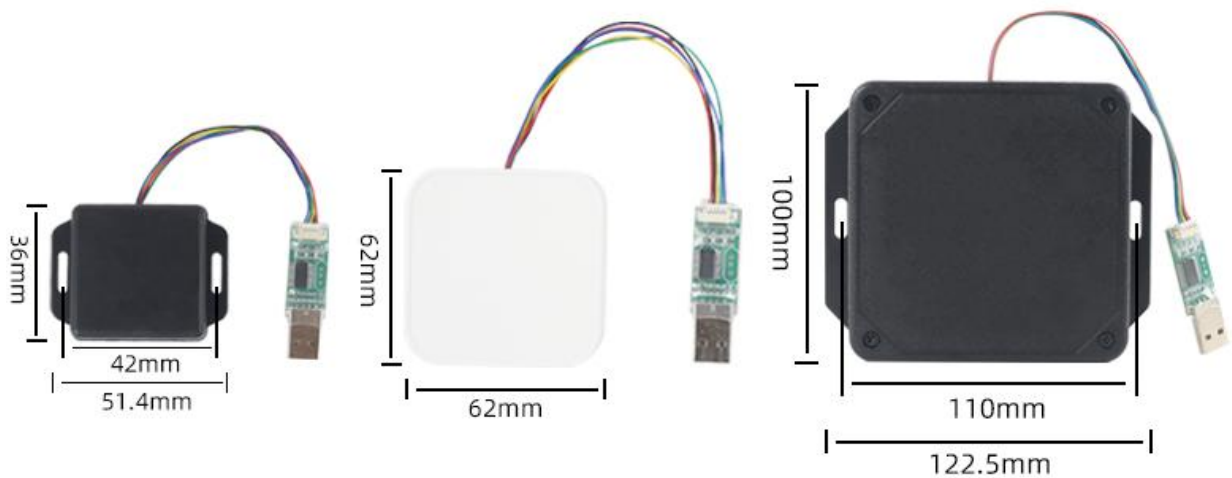


DATASHEET

UHF RFID Reader Module plus for ISO-18000-6C Protocol

FI-50X UHF RFID Reader



Compatible Multiple Platforms



STM/SCM



Android



PC



Raspberry Pi



Arduino



etc....

ETC....

ABSOLUTE MAXIMUM RATINGS

Parameter	Symbol	Conditions
Max storage temperature	T _{STOREmax}	+120°C
Min storage temperature	T _{STOREmin}	-60°C
Power supply voltage	V _{IN}	-0.1V to +5.5V
Electrostatic discharge	V _{ESD}	1KV

Recommend Operating Conditions

Parameter	Symbol	Min.	Typ.	Max.	Unit
Power Supply voltage	V _{IN}	3.6	5	5.5	V
Operating Temperature	T _a	0	-	+50	°C
Storage Temperature	T _s	-40		+85	°C

System Characteristics

Parameter	Description
Interface	TTL3.3V UART
RFID protocol	ISO-18000-6C/ EPC class1 gen2
ANT port impedance	50Ω
Operation range	Around 1.5m, dependent on TAG

ELECTRICAL CHARACTERISTICS

→AC Electrical Characteristics

(Ta = 25 °C, VIN = 5V, VSS = 0V)

Parameter	Symbol	Min.	Typ.	Max.	Unit
RF Output Frequency for reader	F _c	902	-	928	Mhz
RF Output Power	P _{out}	-	-	26.5	dbm
RF Transmission setup time	T _{RF_OUT}	-	-	0.5	ms
RF Frequency error	F _{error}	-	-	1000	ppm
Interrogator Transmit Spurious Emissions, In-Band	In accordance with local regulations				-
Interrogator Transmit Spurious Emissions, Out of-Band	In accordance with local regulations				-
RF Bandwidth	In accordance with local regulations				-
Transmit data rate	T _{Rate}	-	26K	-	bps
Modulation	ASK				
Modulation depth	90% normally				
Data Coding	PIE				
Demodulation	ASK				
Download data rate	D _{Rate}	-	40K	-	bps
Data encoding	FM0				

→DC Electrical Characteristics for Reader mode

(VIN = 5V, VSS = 0V)

Parameter	Symbol	Min.	Typ.	Max.	Unit
Average operating current	I _{OC}	-	280	-	mA
Standby current	I _{SB}	-	-	10	mA
Peak current	I _{peak}	-	300	-	mA
Disable current	I _{dis}	-	0.7	-	uA
Enable pin high (enabled)	V _{EN(HI)}	0.9	-	VIN	V
Enable pin low (disabled)	V _{EN(LO)}	0	-	0.4	V
UART_RX Input Low Voltage	V _{IL}	-0.5	-	0.66	V
UART_RX Input High Voltage	V _{IH}	1.98	-	3.8	V
UART_TX Output Low Voltage	V _{OL}	-	-	0.5	V
UART_TX Output High Voltage	V _{OH}	2.2	-	-	V



Pin	Color	Definition	Instructions
1	Blue	VCC	Ground
2	Green	TXD	Enable power (>1.5V) High(active) Low(Power down),support Not connectable
3	Yellow	RXD	TTL Level(3.3V),output
4	Black	EN	TTL Level(3.3V),input
5	Red	GND	DC 3.7V-5V